

Test Questions

Q 01 Multiple Choice — Conceptual

The fundamental difference between the **pure Chain-Ladder method** and the **Bornhuetter-Ferguson (BF) method** when estimating IBNR is that:

- a) Chain-Ladder uses premium factors, whereas BF uses paid loss factors.
- b) BF balances observed experience with an *a priori* expectation of ultimate losses, giving less weight to the data when they are immature.
- c) Chain-Ladder can only be applied to paid loss triangles, not reported loss triangles.
- d) BF requires more than 10 years of historical experience to be valid.

Answer: b)

The BF method combines two sources of information: the observed experience in the triangle (as does CL) and an *a priori* expectation of ultimate losses (typically derived from pricing or the business plan). The formula for IBNR under BF is:

$$\text{IBNR}_{\text{BF}} = A \text{ priori ultimate} \times \left(1 - \frac{1}{\text{CDF}}\right),$$
$$\text{Ultimate}_{\text{BF}} = \text{Reported losses} + \text{IBNR}_{\text{BF}}.$$

The factor $(1 - 1/\text{CDF})$ represents the proportion of ultimate losses that has not yet been reported. When the accident year is immature (high CDF), this factor is large and the *a priori* estimate dominates; as the year matures ($\text{CDF} \approx 1$), the influence of the *a priori* estimate fades and BF converges to CL. Options a), c), and d) are incorrect: CL can also use reported loss data; BF does not require 10 years of historical experience.

Q 02 Numerical Calculation

The following information is available for **accident year 2023**:

Triangle Data

- Cumulative reported losses (24 months): \$1,200 M.
- 24-to-Ultimate CDF (Chain-Ladder): 2.000.
- *A priori* expected ultimate losses (for BF): \$2,800 M.

Calculate: **(i)** the IBNR reserve using Chain-Ladder and **(ii)** the IBNR reserve using Bornhuetter-Ferguson. Which method produces the higher IBNR, and why?

A 02 Calculation Steps

(i) Chain-Ladder

$$\begin{aligned}\text{Ultimate}_{\text{CL}} &= \$1,200 \text{ M} \times 2.000 = \$2,400 \text{ M}, \\ \text{IBNR}_{\text{CL}} &= \$2,400 \text{ M} - \$1,200 \text{ M} = \$1,200 \text{ M}.\end{aligned}$$

(ii) Bornhuetter-Ferguson

$$\begin{aligned}\text{IBNR}_{\text{BF}} &= \$2,800 \text{ M} \times \left(1 - \frac{1}{2.000}\right) \\ &= \$2,800 \text{ M} \times 0.5000 = \$1,400 \text{ M}, \\ \text{Ultimate}_{\text{BF}} &= \$1,200 \text{ M} + \$1,400 \text{ M} = \$2,600 \text{ M}.\end{aligned}$$

IBNR under BF (**\$1,400 M**) is higher than IBNR under CL (**\$1,200 M**) because the *a priori* expected ultimate of \$2,800 M exceeds the ultimate projected by CL (\$2,400 M). This reflects the fact that losses observed at 24 months have been lower than expected based on pricing; BF places more “trust” in the business plan and sets a more conservative reserve. If the actuary considers the current data more reliable, greater weight can be given to CL.

Q 03 Multiple Choice — Calculation

A loss triangle produces the following **average development factors**:

Selected Factors

$$\begin{aligned}f(12 \rightarrow 24) &= 2.000, \\f(24 \rightarrow 36) &= 1.500, \\f(36 \rightarrow 48) &= 1.200, \\f(48 \rightarrow \text{Ult}) &= 1.050 \quad [\text{tail factor}].\end{aligned}$$

For an accident year with **\$500 M reported at 12 months**, what are the 12-to-Ultimate CDF and the IBNR estimated using Chain-Ladder?

- a) CDF = 4.050 | IBNR = \$1,525 M.
- b) CDF = 5.750 | IBNR = \$2,375 M.
- c) CDF = 3.780 | IBNR = \$1,390 M.
- d) CDF = 4.200 | IBNR = \$1,600 M.

Answer: c) — CDF = 3.780; IBNR = \$1,390 M

$$\begin{aligned}\text{CDF}(12 \rightarrow \text{Ult}) &= 2.000 \times 1.500 \times 1.200 \times 1.050 \\&= 3.000 \times 1.200 \times 1.050 \\&= 3.600 \times 1.050 \\&= 3.780.\end{aligned}$$

$$\begin{aligned}\text{Ultimate} &= \$500 \text{ M} \times 3.780 = \$1,890 \text{ M}, \\ \text{IBNR} &= \$1,890 \text{ M} - \$500 \text{ M} = \$1,390 \text{ M}.\end{aligned}$$

The cumulative development factor is obtained by multiplying the selected development factors from the earliest period to ultimate, **including the tail factor**. Omitting the tail factor (1.050) would yield CDF = 3.600 and IBNR = \$1,300 M, an insufficient reserve. In long-tail lines (liability, occupational risk insurance (ARL), occupational diseases), the tail factor may be the most influential reserving assumption and must be rigorously documented.

Q 04 Short Answer — Conceptual

Explain the difference between **IBNR** (in the strict sense) and **IBNER**. Why does actuarial practice often refer to “broad IBNR” encompassing both concepts, and how does the cumulative loss triangle capture them?

A 04 Conceptual Explanation

IBNR (strict) — Incurred But Not Reported: claims that have already occurred before the valuation date but have not yet been reported to the insurer. There is a time lag between the loss event and its notification; quantification depends on the reporting speed of the line of business.

IBNER — Incurred But Not Enough Reserved: claims already reported and recorded in the system whose case reserves are understated. The final cost will exceed the established reserve due to deterioration, legal fees, or subsequent court rulings.

Broad (“total”) IBNR: in actuarial practice, both are grouped under a single term because the cumulative loss triangle captures the two phenomena *simultaneously*: the development axis includes both late-reported claims (strict IBNR) that appear in future periods and increases in individual case reserves (IBNER). Stochastic methods such as the ODP Bootstrap quantify the combined uncertainty from both sources.

Q 05 Multiple Choice — Colombian Regulation

Under the current Colombian regulatory framework (Decree 2555 of 2010 and SFC circulars), a life insurance company's **claims reserve** must:

- a) Be calculated exclusively using the Chain-Ladder method, as it is the only method validated by the SFC.
- b) Include both reported but unsettled claims (SANL) and an actuarial estimate of incurred but not reported claims (IBNR).
- c) Equal at least 100% of the net premiums earned during the period.
- d) Be calculated only by actuaries with at least 10 years of certified experience.

Answer: b)

Decree 2555 of 2010 (Book 31, Title 4) and SFC instructions require the **claims reserve** to include:

1. **Reported but Unsettled Claims (SANL):** case-by-case reserves for claims being processed, including reported claims under review and those awaiting payment.
2. **IBNR:** an actuarial estimate of incurred but not reported claims. The regulation does not prescribe a specific method; the Responsible Actuary selects and documents the methodology, justifying its suitability to the Statutory Auditor and the SFC in the event of an inspection. Options a), c), and d) are incorrect: the SFC does not mandate CL; the claims reserve is not calculated as a percentage of premiums; and there is no requirement for the actuary to have 10 years of experience (there is, however, a requirement for a certified Responsible Actuary).

Q 06 Multiple Choice — Methodological Judgment

In which of the following scenarios is it **most appropriate** to prefer Bornhuetter-Ferguson over pure Chain-Ladder?

- a) Lines with more than 15 years of homogeneous, stable experience and no changes in business mix.
- b) Very mature accident years (e.g., 72+ months) where most claims are already developed.
- c) Immature accident years (e.g., only 12 months) in long-tail lines with limited internal experience.
- d) When a complete, extensive, and reliable triangle of homogeneous historical data is available.

Answer: c)

Chain-Ladder applies the observed development factors directly to the reported amount. When the data are **immature** (only 12 months, few claims, volatile development factors), CL amplifies statistical error: a factor $f(12 \rightarrow 24)$ estimated with low credibility can drive the projection sharply in either direction.

BF “anchors” the estimate to a reasonable *a priori* expectation (premium $\times$ expected loss ratio from pricing), giving *less* weight to observed data when they are scarce. As the accident year matures and the data gain credibility, the weight of the *a priori* estimate naturally decreases until $BF \approx CL$. Options a), b), and d) describe scenarios where CL is more appropriate because the data are mature and reliable, making an external *a priori* estimate unnecessary.

Q 07 Short Answer — Advanced

Describe the **Cape Cod method** (Stanard-Bühlmann) and explain how it differs from classical BF. When is one preferable to the other? Include the formula for estimating the ELR under Cape Cod.

Answer

The **Cape Cod method** (Stanard-Bühlmann) estimates the *a priori* expected loss ratio (ELR) directly from the triangle's own data, rather than setting it externally as in classical BF:

$$\text{ELR}_{\text{CC}} = \frac{\sum_i \text{Reported losses}_i}{\sum_i (\text{Premium}_i \times \% \text{ developed}_i)}.$$

Where

$$\% \text{ developed}_i = \frac{1}{\text{CDF}_i}.$$

$$\text{IBNR}_{\text{CC},i} = \text{ELR}_{\text{CC}} \times \text{Premium}_i \times \left(1 - \frac{1}{\text{CDF}_i}\right).$$

Key Difference from Classical BF:

In BF, the ELR is set externally (from pricing or the business plan). It is more subjective but stable when data are scarce. In Cape Cod, the ELR emerges from observed data weighted by development; it is more objective but can be volatile with only a few years of experience.

When should each method be preferred? Cape Cod is used when no reliable external ELR is available or when internal consistency across accident years is sought; BF is used when the pricing ELR is credible, the line of business is new, or data are very scarce. In practice, comparing both methods alongside CL forms the core of reserve analysis.